TELMO MONTEIRO

Fundação da Ciência e Tecnologia (FCT) Doctoral Fellow at IA/CAUP, Portugal
telmo.monteiro@astro.up.pt ◇ ORCID: 0000-0001-8991-4615 ◇ https://telmonteiro.github.io

EDUCATION

Faculty of Sciences of University of Porto (FCUP) September 2025 - Present
Astronomy, Ph.D. *Porto, Portugal*

- Thesis: "Planets as probes of stellar atmospheres: Characterizing local spectra via Doppler Shadow"
- Supervisors: Eduardo Cristo (IA/CAUP), William Dethier (IA/CAUP), Nuno C. Santos (IA/CAUP, FCUP)

Faculty of Sciences of University of Porto (FCUP) September 2023 - July 2025
Astronomy and Astrophysics (Technologies for Space Sciences), M.Sc. *Porto, Portugal*

- Dissertation: "Analysis of NIR activity indices for exoplanet detection and characterization"
- Supervisors: João G. da Silva (IA/CAUP), Elisa Delago-Mena (CAB-INTA), Nuno C. Santos (IA/CAUP, FCUP)

University of Aveiro September 2019 - June 2023
Physics, Licenciatura (Portuguese 1st cycle) *Aveiro, Portugal*

- Dissertation: "Effective temperature determination methods of FGK stars"
- Supervisors: João Fernandes (UC, CITEUC), Vítor Bonifácio (UA)

RESEARCH EXPERIENCE

Institute for Astrophysics and Space Sciences (IA/CAUP) - FCUP September 2025 - Present
Doctoral Researcher *Porto, Portugal*

Institute for Astrophysics and Space Sciences (IA/CAUP) - FCUP February - July 2024
Extracurricular intern via PEEC *Porto, Portugal*

Institute for Astrophysics and Space Sciences (IA/CAUP) July 2022
Summer Intern *Portugal*

Institute for Astrophysics and Space Sciences (IA/CAUP) July 2021
Summer Intern *Portugal*

IT - Telecommunications Institute February – August 2021
Research Initiation Fellow *Aveiro,Portugal*

PUBLICATIONS

First author:

1. **Monteiro T.**, Gomes da Silva J., Delgado-Mena E., Santos N. C., et al. "Sensitivity of single-line NIR activity indices in cool stars using HARPS and NIRPS simultaneous observations", Astronomy & Astrophysics, under revisions

Co-author:

1. Genest F., Lafrenière D., Allart R., Pelletier S., et al. (incl. **Monteiro T.**) "Refined Atmospheric Constraints for WASP-80 b: Joint Analysis of NIRPS and JWST/NIRCam Transmission Spectra", Astronomy & Astrophysics, under revisions
2. Al Moulla K., Dumusque X., Faria J., Santos N. C., et al. (incl. **Monteiro T.**) "The Sun Through Rose-Colored Glasses. Three Years of Near-Infrared Sun-as-Star Precision Spectroscopy with NIRPS", Astronomy & Astrophysics, under revisions
3. Larue P., Delfosse X., Mignon L., Hara N., et al. (incl. **Monteiro T.**) "Revisiting Gl581 with NIRPS and HARPS. Stellar activity investigation, updated orbital parameters, and improved planet detection limits", Astronomy & Astrophysics Journal, under review
4. Weisserman D., Gromek N., Cloutier R., Bali K., et al. (incl. **Monteiro T.**) "Super-Earth masses and stellar abundances from NIRPS reveal tentative evidence for water-rich formation around M dwarfs", Astronomy & Astrophysics Journal, Volume 709, A165. (2026)

5. Gomes da Silva J., Delgado-Mena E., Santos N. C., **Monteiro T.**, et al. "Blind search for activity-sensitive lines in the near-infrared using HARPS and NIRPS observations of Proxima and Gl 581", Astronomy & Astrophysics Journal, Volume 700, A177. (2025)
6. Gomes da Silva J., Bensabat A., **Monteiro T.**, Santos N. C. "Optimising the Hα index for the identification of activity signals in FGK stars", Astronomy & Astrophysics Journal, Volume 668, A174. (2022)
7. Salgueiro G., **Monteiro T.**, Costa D., Teixeira, M., Pinto A. N., Santos J. P. "High quality random number generator from constrained devices using post processing methods applied to a high level network", Communications and Quantum Computing Workshop (WQUANTUM) Proceedings, p. 19-24. (2021)

GRANTS AND AWARDS

Total: ∼149,000€

HORS3 Conference Travel Grant (April 2026) - 280€ - Accepted

COST Action "PLANETS" Inclusiveness Target Country (ITC) Conference Grant (April 2026) - 1,055€ - Accepted

Cool Stars 23 Conference Fee Waiver (February 2026) - 270€ - Accepted

Exploring Star-Planet-Disk Connections Conference Travel Grant (June 2025) - 300€ - Accepted

Fundação da Ciência e Tecnologia (FCT) Ph.D. Studentship (August 2025) - ≥85,900€ - Accepted

Erasmus Mundus Joint Master (EMJM) MASS scholarship (September 2023) - 58,600€ - Refused

Research Initiation Grant at Telecommunications Institute (January 2021) - 2,600€ - Accepted

TECHNICAL STRENGTHS

Programming Languages	Python, MATLAB, SQLite, R, LaTeX
Languages	Portuguese (Native), English (C1 level certificate by Universidade de Aveiro)

OBSERVING EXPERIENCE

Paranal Observatory, Chile (remote) - PoET solar telescope, 10 days (2026)

MEMBERSHIP

PoET Consortium Science Team Member (2025 - Present)

NIRPS Consortium Science Team Member (2024 - Present)

IA Planetary Systems Team Member (2024 - Present)

CONFERENCES, WORKSHOPS AND MEETINGS

Contributed Talks:

"A Marauder's Map for stellar surfaces via transiting exoplanets" - XXXVI Astronomy and Astrophysics National Meeting (September 2026, Porto, Portugal)

"Resolving local stellar spectra for more reliable transmission spectroscopy" - High-Resolution Exoplanet and Stellar Characterization Today and in the ELT Era (HORS3S) (July 2026, Granada, Spain)

"Analysis of NIR activity indices for exoplanet detection and characterization" - NIRPS Science Team Meeting (November 2025, Online)

"Analysis of NIR activity indices for exoplanet detection and characterization" - Exploring Star-Planet-Disk Connections Conference (October 2025, Tartu, Estonia)

"Analysis of NIR activity indices for exoplanet detection and characterization" - University of Porto's Young Research Meeting IJUP (May 2025, Porto, Portugal)

Posters:

'Using transiting exoplanets to resolve local stellar surface" - Exoplanets 6 (June 2026, Porto, Portugal)

"Planets as probes of stellar atmospheres: Characterizing local spectra via Doppler Shadow" - Cool Stars 23 (June 2026, Tokyo, Japan)

"Analysis of NIR activity indices for exoplanet detection and characterization" - XXXV Astronomy and Astrophysics National Meeting (September 2025, Lisbon, Portugal)

"Analysis of NIR activity indices for exoplanet detection and characterization" - Sagan Summer Workshop 2025 (July 2025, Pasadena, CA, USA)

"Analysis of NIR activity indices for exoplanet detection and characterization" - The 6th Workshop on Extremely Precise Radial Velocities (July 2025, Porto, Portugal)

"Stellar Activity in Stars with Exoplanets: Presenting the SAITAMA pipeline" - XXXIV Astronomy and Astrophysics National Meeting (September 2024, Guimarães, Portugal)

Member of Local Organizing Committee (LOC):

NIRPS Science Team Meeting (November 2026, Porto, Portugal)

Exoplanets 6 (June 2026, Porto, Portugal)

Attendee:

NIRPS Science Team Meeting (May 2025, Online)

NIRPS Science Team Meeting (November 2024, Grenoble, France – online attendee)

PoET Workshop #2 (October 2024, Porto, Portugal)

XXXIII Astronomy and Astrophysics National Meeting (September 2023, Coimbra, Portugal)

ASSOCIATIVISM AND VOLUNTEERING

University of Porto — October 2026 – Today
M.Sc. and Ph.D. Students representative on the CAUP Board+Services meetings — *Porto, Portugal*

Physis - Portuguese Association of Physics Students — Portugal
- Secretary-Archivist of the General Assembly Board (November 2025 - November 2026)
- Coordinator of the Career Prospects Departament (November 2024 – November 2025)
- Pedagogic Departament Member (February 2021 – November 2021)

University of Porto — November 2023 – Today
Students representative of the M.Sc. in A&A Course Monitoring Committee — *Porto, Portugal*

Physics Department - Universidade de Aveiro — September 2022 – July 2023
Mentor in the PeerTeaching programme in Classical Mechanics and MATLAB — *Aveiro, Portugal*

Physics Students National Meeting 2023 Organizing Committee (CO-ENEF) — July 2022 – March 2023
General Coordinator — *Aveiro, Portugal*

EPS Young Minds Aveiro — October 2020 – July 2022
Pedagogic Department Member — *Aveiro, Portugal*

University of Aveiro — October 2020 – August 2022
Students representative in the Licenciatura in Physics Course Committee — *Aveiro, Portugal*

OUTREACH

Outreach presentation "Mapping the surface of the stars" — 18 september 2026
Ignite IAstro — *Armamar, Portugal*

IA and Porto Planetarium Science Communicator — 23 - 26 april 2026
Comic Con Portugal — *Santa Maria da Feira, Portugal*

DFA Eureka room guide — February 2026 - Today
Physics and Astronomy Department of Universidade do Porto — *Porto, Portugal*

Physics Days - roundtable MSc and PhD students — April 2025
Physics and Astronomy Department of Universidade do Porto — *Porto, Portugal*

Mostra UP — March 2025
Universidade do Porto

Volunteer at the open university fair, representing the Physics and Astronomy Department.

"Physis Masters" May 2024
Physis - Portuguese Association of Physics Students *Portugal*
- Talked about my experience in the Master's degree in Astronomy and Astrophysics at FCUP.

"Projects and Theses Roundtable" November 2022
EPS Young Minds Aveiro *Aveiro, Portugal*
- Talked about my Licenciatura's project and my experience in making it.

Engineering Physics vs Physics: similarities, differences and what you can find in UAveiro June 2022
Uniarea *Aveiro, Portugal*
- Wrote about the Licenciatura in Physics at Universidade de Aveiro for a promotional article, as well participated in an Instagram live session.

Licenciatura in Physics at Universidade de Aveiro November 2020
Inspiring Future *Aveiro, Portugal*
- Talked about the Licenciatura in a promotional video.